From Data to Prevention:
Predicting Bone Fractures in Diabetic Men Over 60

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Texas Undergraduate Mathematics Conference
October 18, 2025

Importance of Risk Assessment

- Fracture risk assessment is a crucial component of effective **healthcare management**.

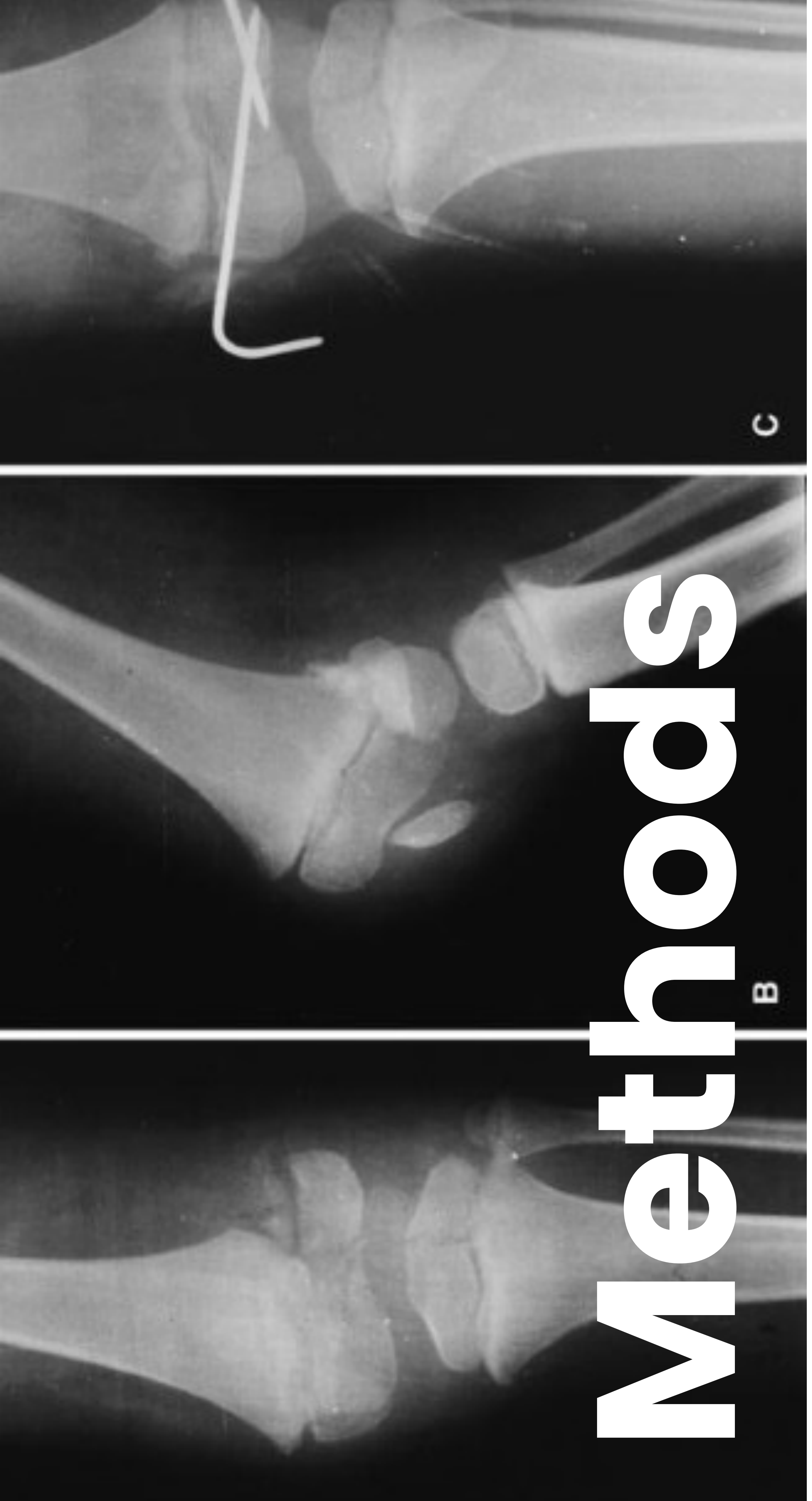
- Chronic Diabetes Mellitus (DM) contributes to glucose toxicity due to hyperglycemia, leading to:
  - Decreased bone formation
  - Altered or reduced bone turnover
  - **Increased risk of fractures**

- The relationship between DM and fracture susceptibility is **underexplored in the Hispanic male population.**

Context & Need for Study

- Hispanic residents are now the largest ethnic group in Texas.
- In South Texas, the **prevalence of DM exceeds 30% in individuals over 40.**
- With the growing Hispanic population and high DM rates in the U.S., understanding fracture risk is vital.

Study Objective:
To compare fracture prevalence between diabetic and non-diabetic men over 60 in San Antonio, Texas, using electronic medical records.



Methods

•Retrospective analysis of a de-identified dataset containing **26,510 male patients** of a local healthcare practice **between 2012 and 2022**. No identifiers are included in the dataset. This study has an exempt IRB status (UIW IRB).

•International Classification of Diseases (ICD) 9/10 codes were used to identify **major osteoporotic fracture (MOF) types**, including **proximal and distal femoral fractures** (shaft fractures were excluded), **proximal humerus fractures**, and **wrist fractures.**

•Vertebral fractures and any other types such as traumatic or pathological fractures were excluded.

•We included the following variables: **age, ethnicity/race(assigned ethnicity), HbA1c, and duration of DM.**

Analysis

We calculated **descriptive statistics** to compare fracture prevalence between **diabetic and non-diabetic men**, stratified by race/ethnicity(assigned ethnicity).

We performed **logistic regression analyses** to examine the associations between:

1) Major osteoporotic fractures (MOF) and the factors: age, HbA1c levels, and duration of diabetes.

2) Femur fractures and the same factors

Software: All statistical analyses were performed using **Python** with the statsmodels library for logistic regression and **Pandas** for data manipulation.

Data Cleaning

Initial Profiling (Target Population)

- Columns: **anon_id, member_gender, member_age, member_race, Member_Ethnicity, market**
  - member_gender (all male) & market (all San Antonio) → drop
- Age: min 60, mean ≈ 72.26, max 89.
- member_race had **22 variants** (case/synonym inconsistencies).

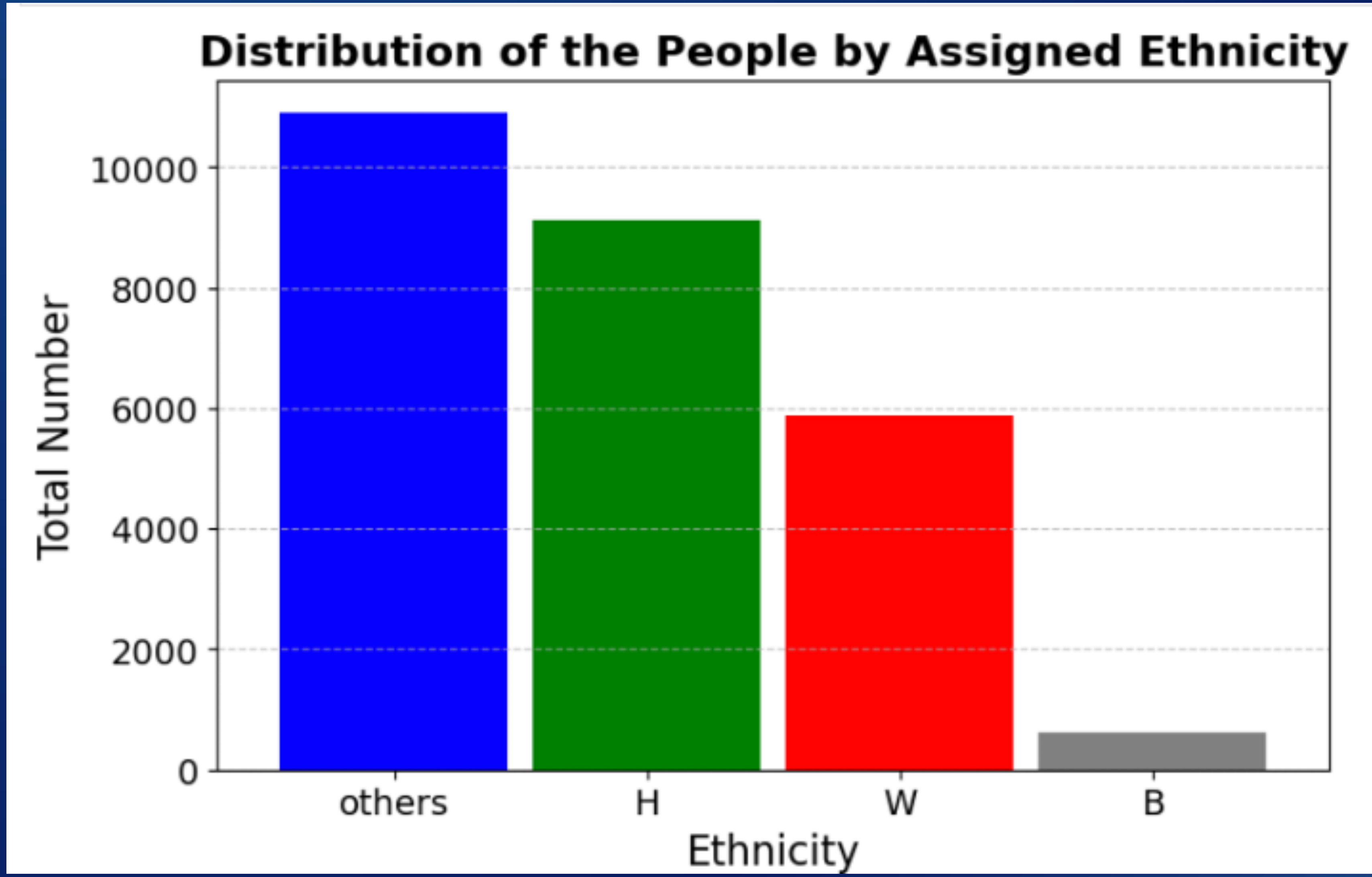
Race Normalization: Consolidated aliases/case to standard labels

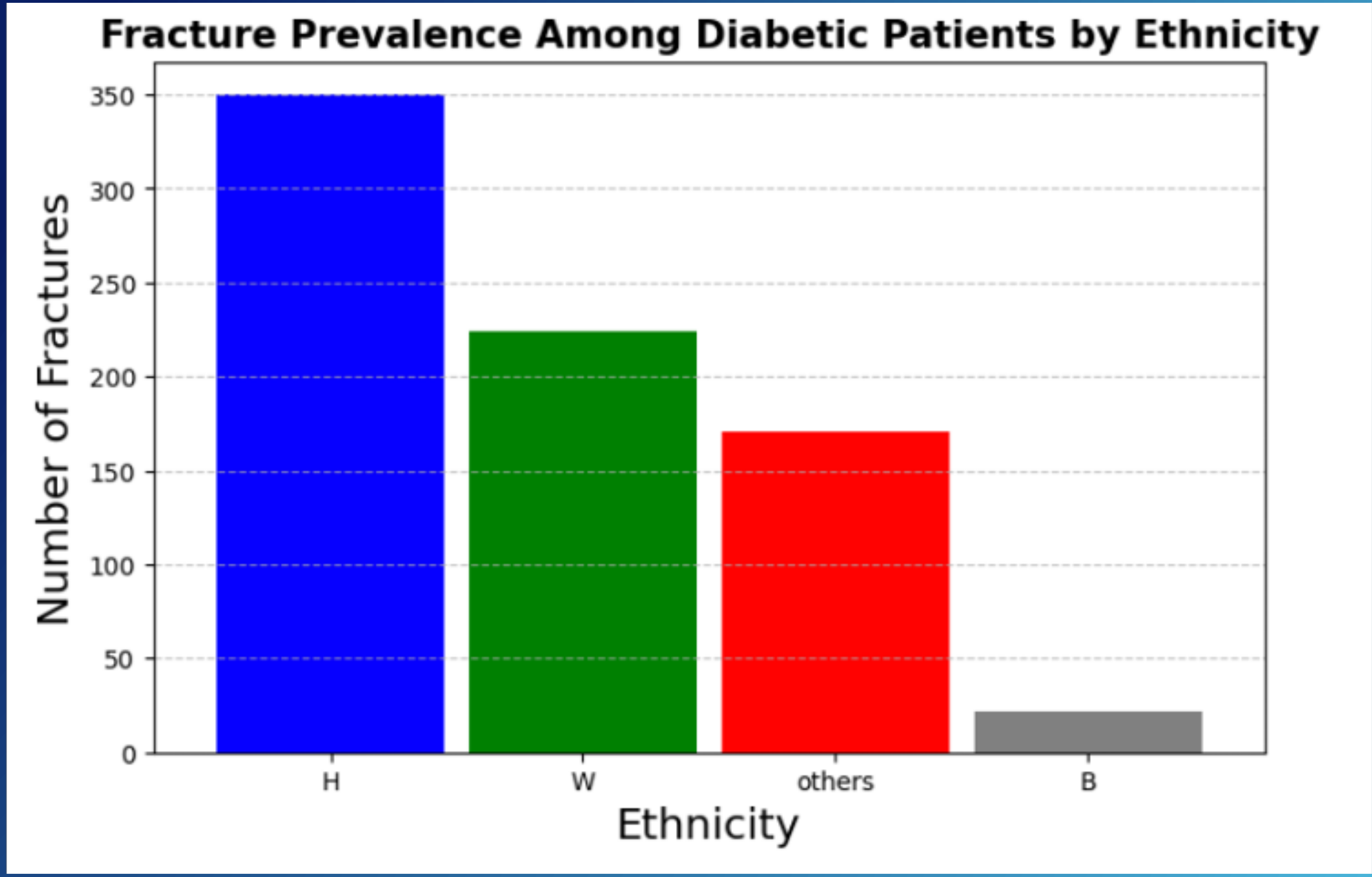
- White, Caucasian, European, English → **White (W)**
- African American, Black → **Black (B)**
- Declined to Specify, Unreported/Refused to Report, Declined To Specify → **Others**
- American Indian, White Earth, Spanish American Indian, Mexican American Indian → **Others**

Data Cleaning: Assigned Ethnicity

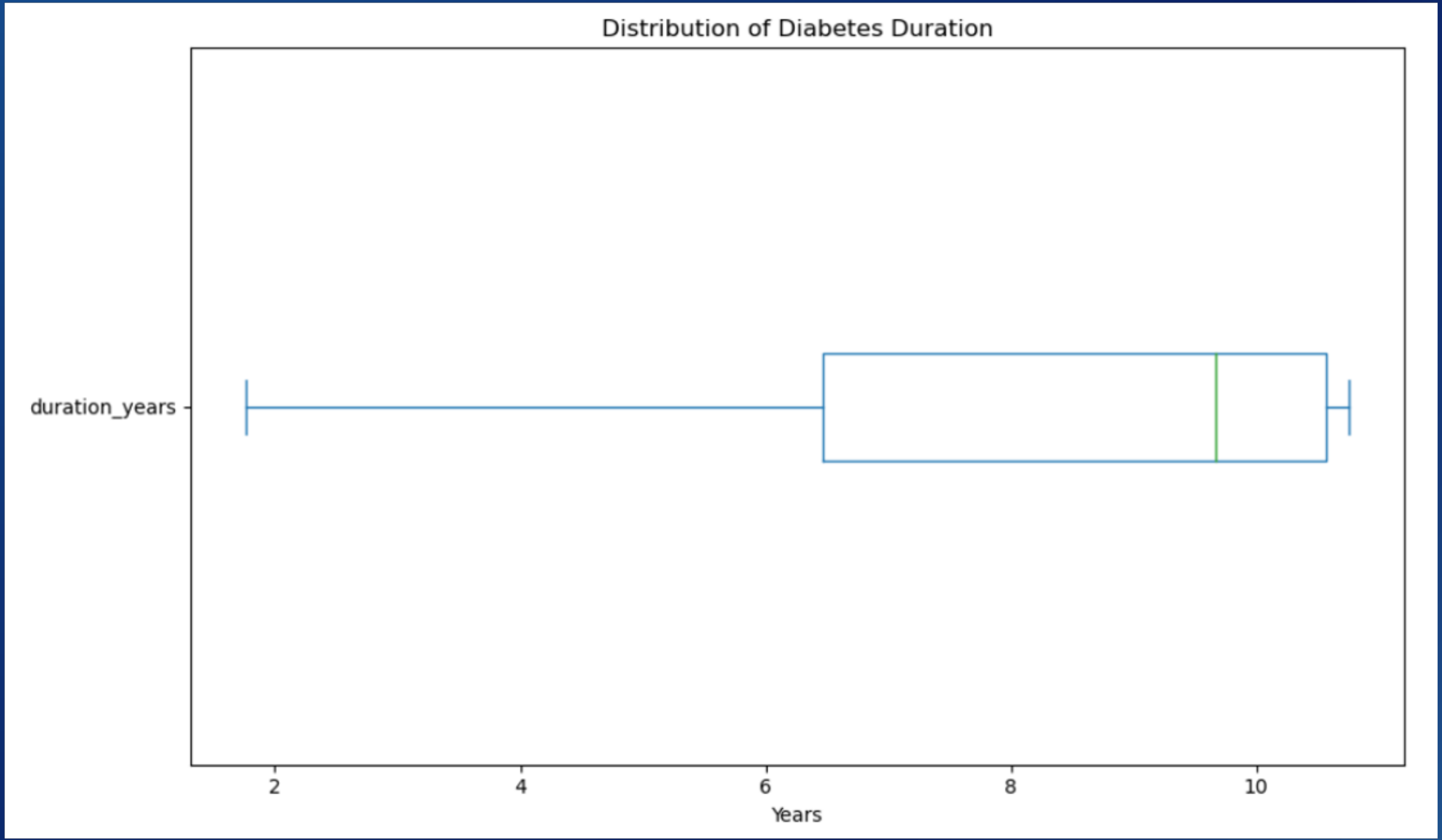
Rules applied to assign consistent categories:

White + Hispanic → **Hispanic (H)**
White + Not Hispanic → **White (W)**
Black + Others → **Black (B)**
Others / Unreported → **Others**





Duration of Diabetes in the Cohort



•Total diabetic patients: 8,682 (32.7% of population)

•**Mean duration: 8.3 years**, Median duration: 9.7 years, Range: 1.7 – 10.7 years, IQR: ~4 years

•**75% of patients had diabetes ≥6.5 years**

Results

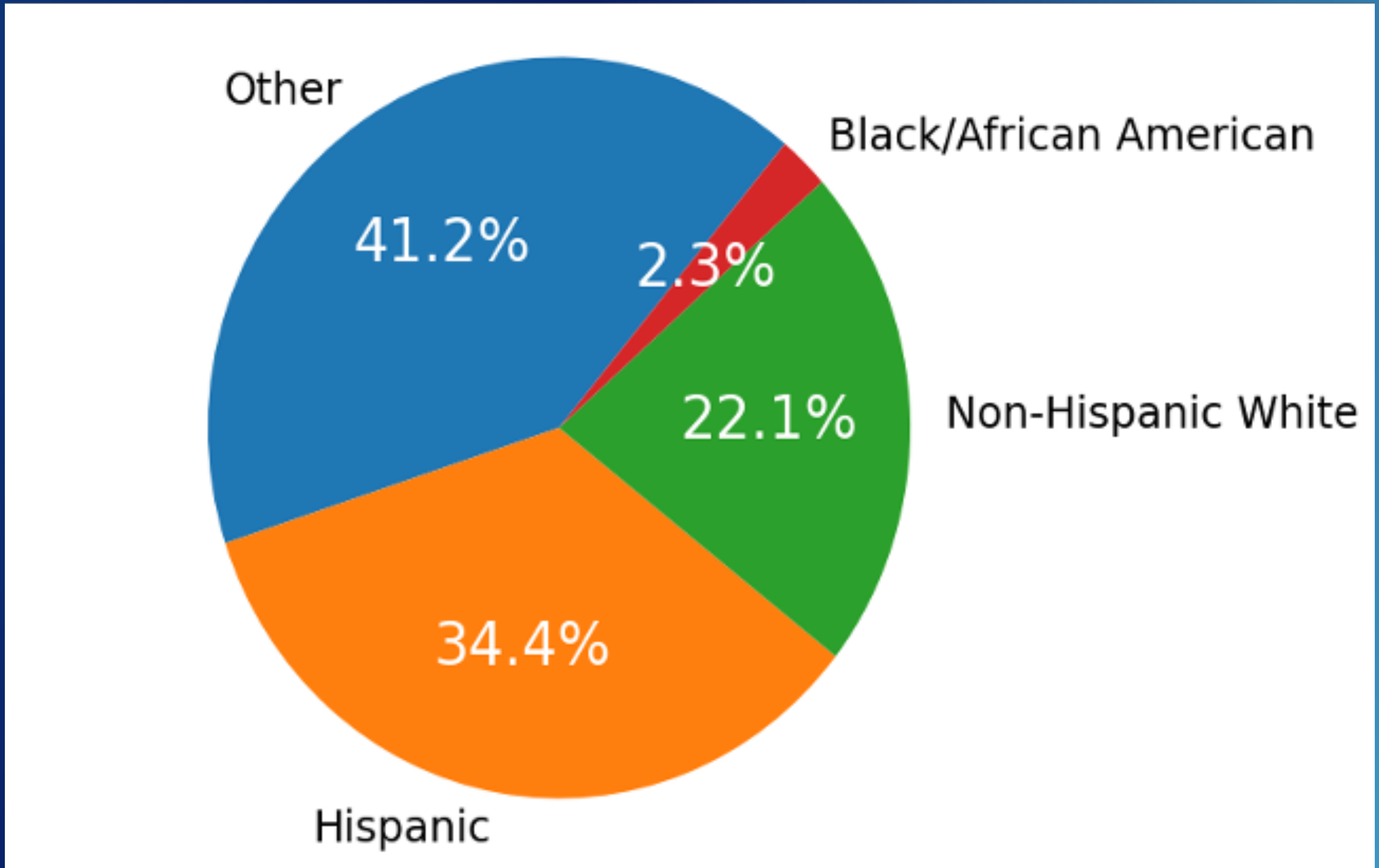


Figure 1. Pie chart indicating the **race/ethnicity representation** in the sample. This all-male sample from San Antonio, TX is made up of **22.1% non-Hispanic White, 34.4% Hispanic, 2.3% Black/African American, and 41.2% Other** (self-identified and unreported categories) patients.

Results

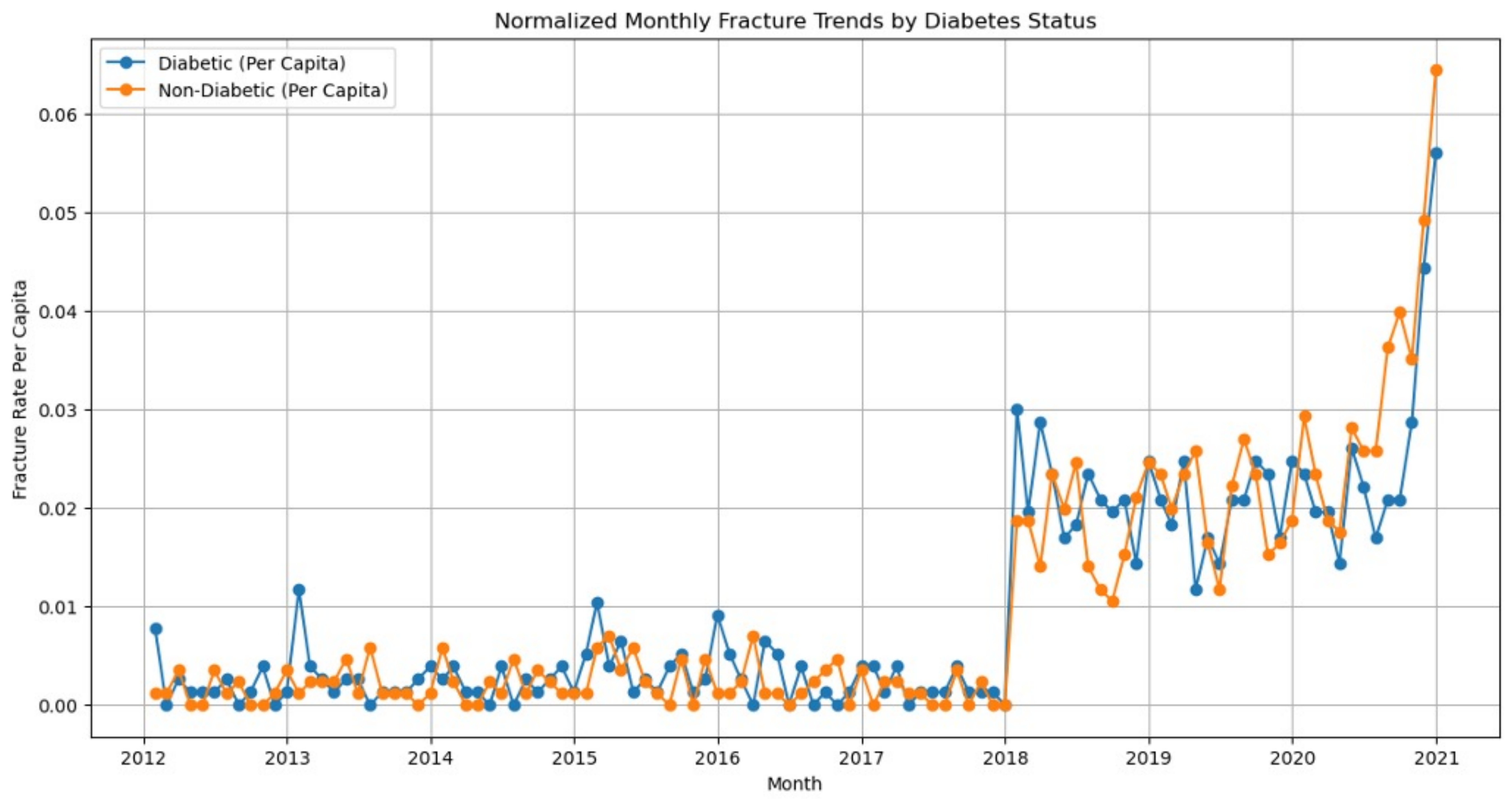
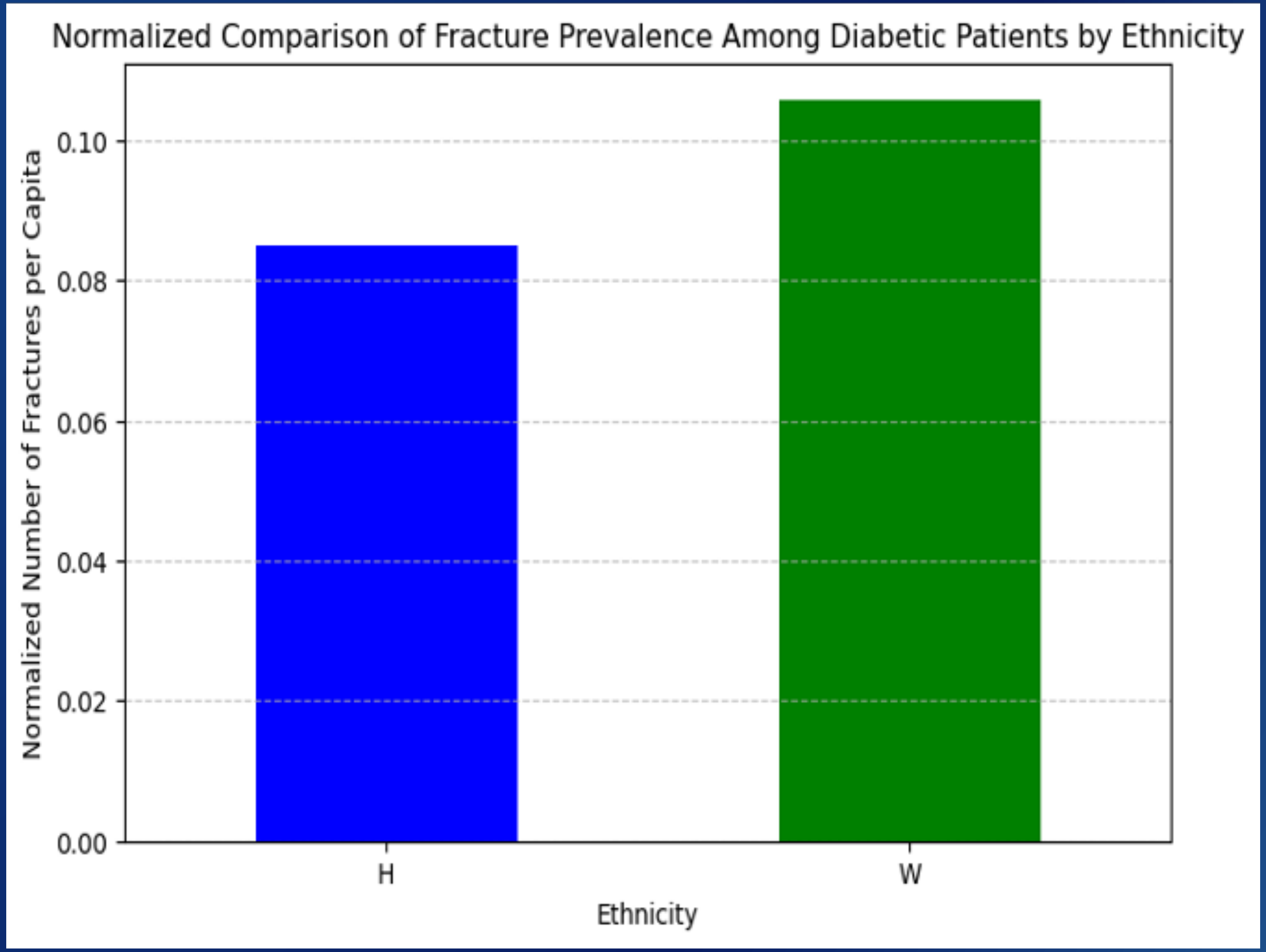


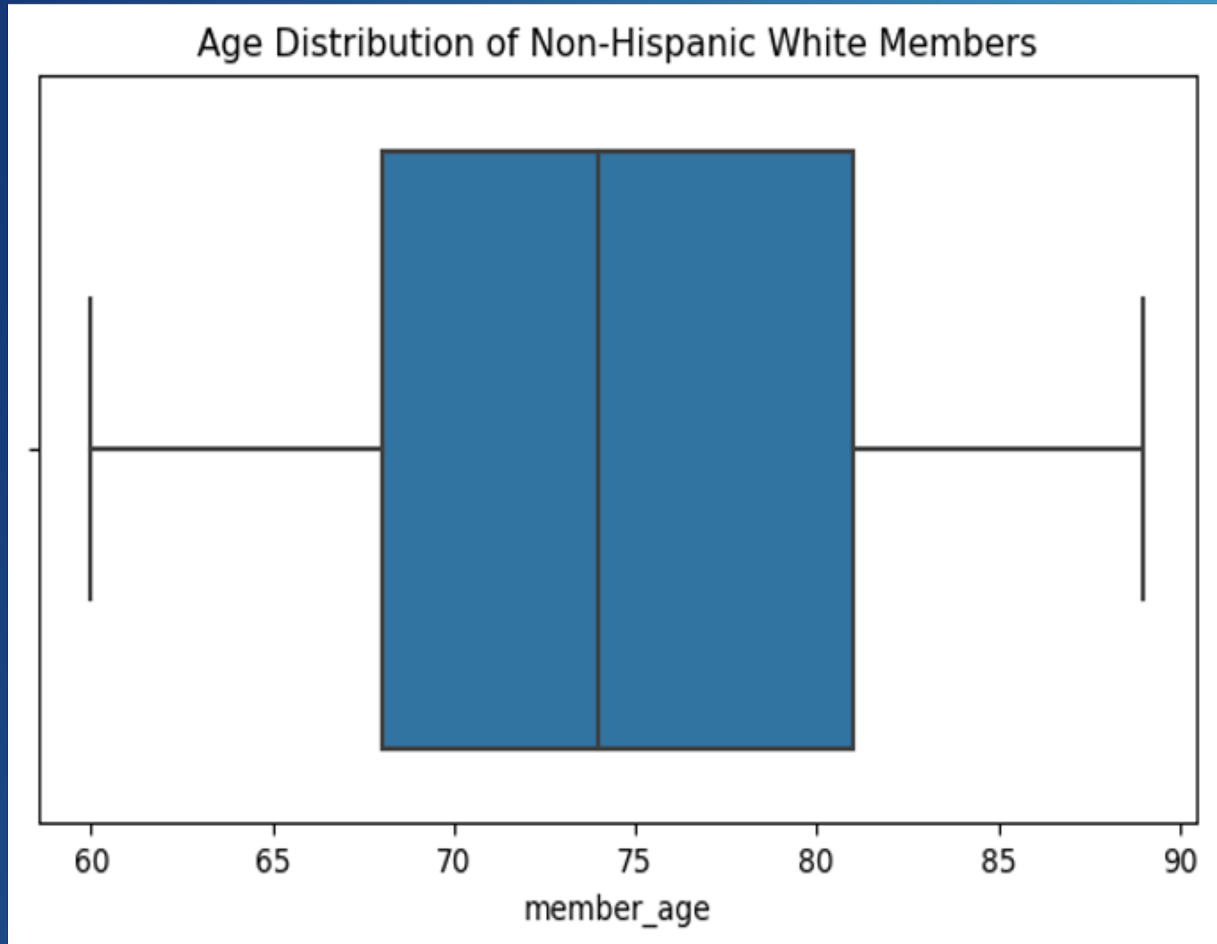
Figure 2. The plot displays the normalized monthly fracture trends per capita for diabetic versus non-diabetic patients.

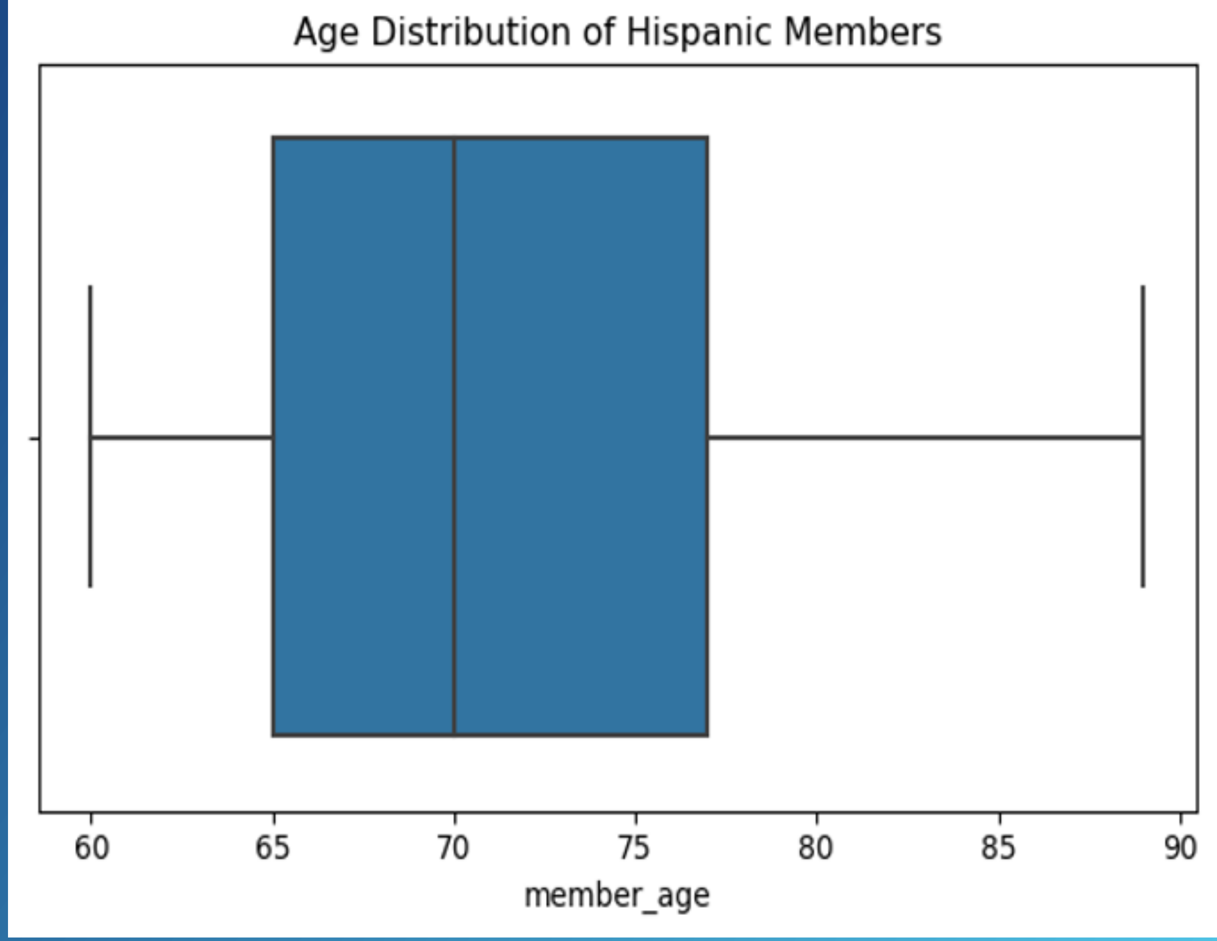
Results



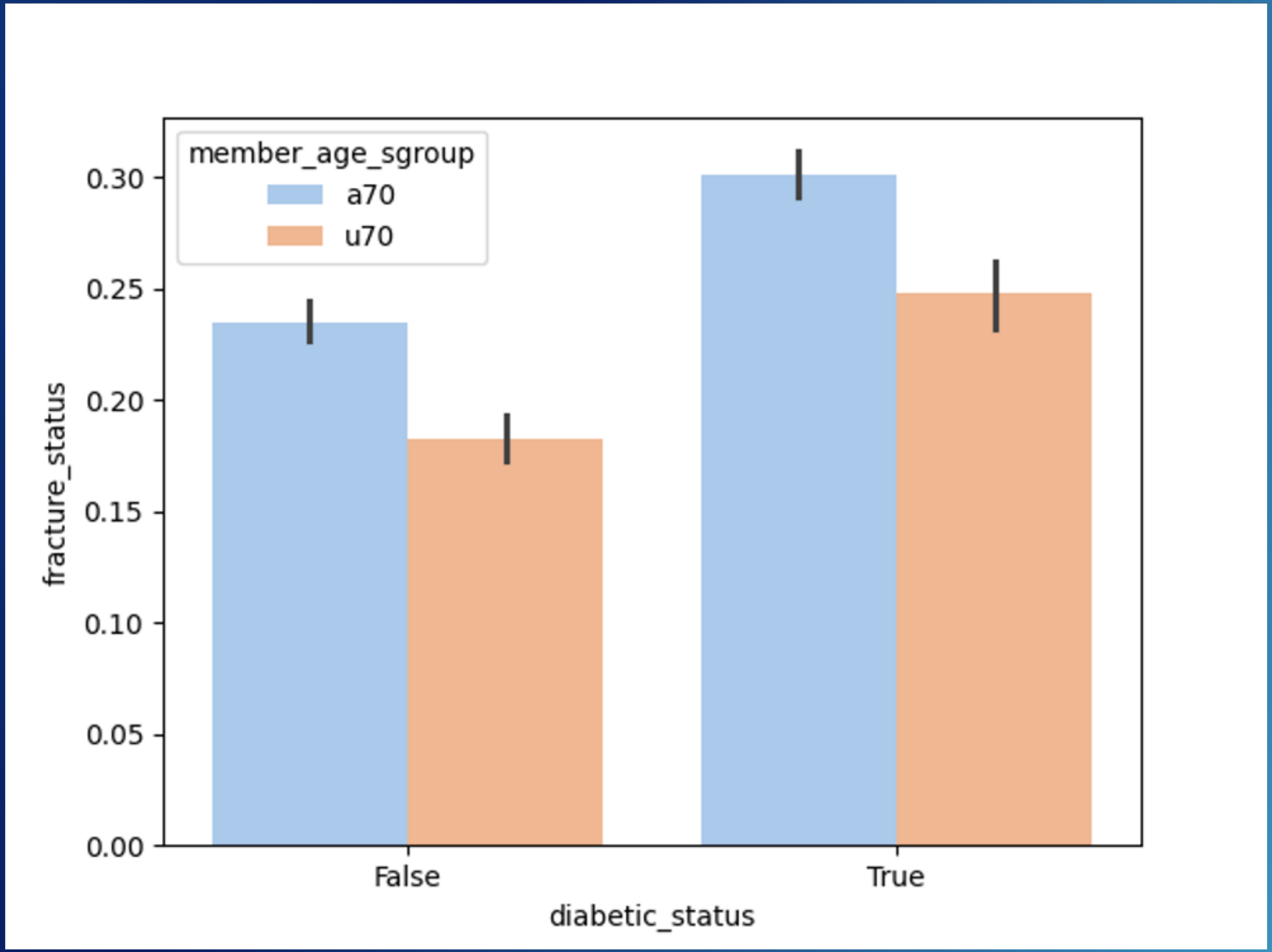
Non-Hispanic White men have a higher normalized fracture prevalence than Hispanic men (about 0.105 vs 0.085).

Non-Hispanic White men show an older age profile.





Results



Fracture rates are **consistently higher in individuals over 70**, especially among diabetics.
This supports the hypothesis that both **age and diabetes increase fracture risk.**

Table 1. Descriptors of the patient sample, by ethnicity/race, age, and fracture counts

Ethnicity/Race	Mean Age (SD)	Sample Size (N)	Major Osteoporotic Fractures (N)	Femur Fractures (N)
Non-Hispanic White	74.4 (7.8)	5864	115	31
Hispanic	71.6 (7.4)	9115	123	38
Black/African American	71.8 (7.6)	621	6	1
Other	71.7 (7.9)	10910	91	23

Table 2. Diabetes characteristics of the sample.

Ethnicity/Race	DM Prevalence (%)	Avg HbA1c (SD)	Age of DM Diagnosis	Duration of DM
Non-Hispanic White	36.10%	6.16 (1.03)	66.66 (7.06)	8.75 (2.49)
Hispanic	45.20%	6.64 (1.49)	64.13 (6.74)	8.3 (2.75)
Black/African American	46.10%	6.39 (1.19)	64.32 (6.86)	8.44 (2.7)
Other	19.80%	6.5 (1.39)	64.57 (6.96)	7.79 (3.05)

- Hispanics and non-Hispanic Whites have comparable major osteoporotic fracture counts (123 vs. 115).
- Black/African Americans (46.1%) and Hispanics (45.2%) show the highest diabetes rates.
- Non-Hispanic Whites have the highest mean age (74.4 years) and later diabetes diagnosis (~66 years).

Table 3. Most common fracture diagnoses in the dataset.

Diagnosis Code	Location	Fracture Count
M81.0	Unspecified osteoporotic (nonpathologic)	217
S72.00*A	Femoral neck (left)	37
S72.14*A	Femoral trochanter	12

Most Frequent Fracture Diagnosis Codes:

- `M81.0` (Osteoporosis without current pathological fracture) is the most frequently occurring diagnosis with **217** instances.
- `S72.002A` (Fracture of unspecified part of neck of left femur, initial encounter for closed fracture) appears **23** times.
- `S72.001A` (Fracture of unspecified part of neck of right femur, initial encounter for closed fracture) is noted **14** times.
- Other codes such as `S72.141A`, `S72.142A`, and `S72.009A` appear less frequently but still significantly across the dataset.

Predictors of Fracture Risk: Logistic Regression Results

Table 4. Logistic regression results: predictors of fracture risk using age, HbA1c, and duration of DM in this cohort.

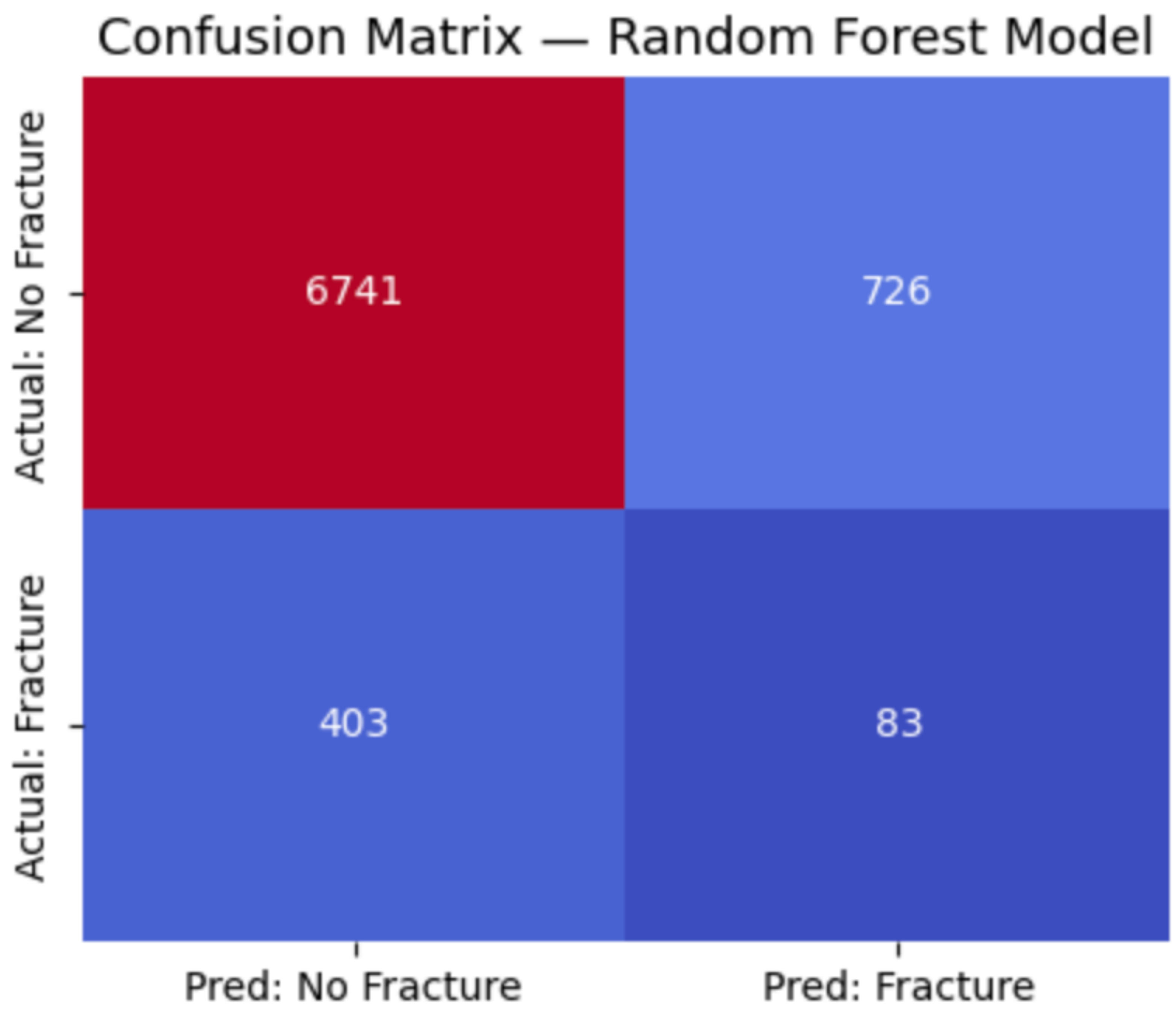
Dependent Variable	Independent Variable	Coefficient (β)	Standard Error	p-value	Odds Ratio (OR)	95% CI (OR)
MOF	Intercept	-6.2868	0.768	0		
	Age	0.0524	0.008	0	1.054	1.037 to 1.068
	HbA1c	-0.2994	0.074	0	0.741	0.555 to 0.846
	Duration of Diabetes	0.0403	0.014	0.004	1.041	1.013 to 1.068
Femur Fractures	Intercept	-12.5333	1.36	0		
	Age	0.0901	0.014	0	1.094	1.062 to 1.118
	HbA1c	-0.0341	0.106	0.749	0.966	0.757 to 1.189
	Duration of Diabetes	0.0846	0.023	0	1.088	1.039 to 1.130

- Logistic regression analysis identified significant predictors of fracture risk in the cohort.
- Age was a strong predictor for both MOF and femur fractures, with odds ratios indicating increased risk per year.
- Duration of diabetes was positively associated with fracture risk, particularly significant for femur fractures.
- HbA1c showed a protective effect in MOF but was not significant for femur fractures.

Random Forest Model

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.90	0.92	7467
1	0.10	0.17	0.13	486
accuracy			0.86	7953
macro avg	0.52	0.54	0.53	7953
weighted avg	0.89	0.86	0.87	7953



Key Takeaways

Overview: Two models were developed to predict bone fractures in diabetic men over 60 using age, years with diabetes, and HbA1c levels.

Random Forest Model:

- Achieved higher overall accuracy (86%) and stronger balance between sensitivity and precision.
- Correctly identified most non-fracture cases while still detecting meaningful fracture risk patterns.
- Better suited for clinical or predictive applications where reliable, data-driven decisions are key.

The logistic model acts as a wide safety net for early detection, while the Random Forest serves as a more dependable diagnostic tool. Together, they demonstrate the trade-off between sensitivity and accuracy in predictive modeling.

Discussion

- This study investigated the association between **DM and low-energy fractures** in a large south Texas male cohort.

- **Duration of DM** was a significant factor for fractures, aligning with previous research.

- Unexpectedly, **HbA1c was a negative predictor** for MOF and not significant for femur fractures, indicating the need for further exploration.

Limitations

•Lack of DEXA scan data limits bone density analysis.

•Comorbidity data, including insulin and glucocorticoid use, were not included.

•Absence of **lifestyle factors**, such as physical activity and dietary habits, in the analysis.

Conclusion

- The results show that **DM increases the odds of fracture in males, but overall fracture prevalence is low**.
- Findings partially align with previous research; yet more investigation is needed to fully understand DM's impact on bone health.
- Continued research is essential to integrate more comprehensive data and to draw clearer connections between **DM, bone metabolism, and fracture risks in males**.

Future Work

- Plan to incorporate DEXA scan data and comorbidity information.
- Analyze the impact of insulin and glucocorticoid use on fracture risk.
- Investigate further to understand the unexpected HbA1c findings.
- Other machine learning algorithms like XGBoost

Special Thanks to our Co-Authors!

- **Shivani Akula, Janay Rocha, Jaydee Foster**
  UIW School of Osteopathic Medicine

- **Michael Almaleh, M.D.**
  WellMed Medical Group

- **Ellen Quillen, Ph.D.**
  Wake Forest University School of Medicine

- **Todd Bredbenner, Ph.D.**
  University of Colorado Colorado Springs

- **Roberto J. Fajardo, Ph.D.**
  UIW School of Osteopathic Medicine

Thank you!